

Bahri Batuhan Bilecen

Doctoral researcher working in generative modeling and representation learning at ETH Zürich.

bbatuhanbilecen@gmail.com ✉
scholar page (4GUUoH4AAAA) G
linkedin.com/in/bbatuhan in
github.com/three-bee O

Education

- **ETH Zürich** Zürich, Switzerland
PhD in Computer Science 09/2025
- **Bilkent University** Ankara, Turkey
MSc in Computer Engineering, with high honors 09/2022 - 05/2025
 - Thesis: Three-Dimensional Reconstruction and Editing from Single Images with Generative Models
 - Supervisors: Aysegul Dundar, Ning Yu
- **Middle East Technical University (METU)** Ankara, Turkey
BSc in Electrical and Electronics Engineering, with high honors 09/2018 - 08/2022
 - Thesis: Comparison and Evaluation of Frame-based Optical Flow and Event-based Optical Flow
 - Supervisors: A. Aydin Alatan

Experience

- **Graduate Researcher** Zurich, Switzerland
VLG @ ETH Zürich 09/2025
 - Working on representation learning and controllable generative models, supervised by Prof. Siyu Tang, Prof. Bernt Schiele, and Dr. Jan Eric Lenssen.
 - Currently in collaboration with Dr. Vassilis Choutas, and Dr. Thabo Beeler, under Google XR and ETHAR.
- **Graduate Researcher** Ankara, Turkey
DLR @ Bilkent University 09/2022 - 09/2025
 - Focused on 2D & 3D generative networks, under the supervision of Prof. Aysegul Dundar.
 - Mentored undergraduate and graduate students, graded assignments, and proctored exams. Reviewed manuscripts submitted to *IEEE TIP, ICML, ECCV, CVPR, ICCV, and NeurIPS*.
- **Research Engineer** Ankara, Turkey
ASELSAN Research Branch 11/2021 - 01/2025
 - Focused on inverse problems in image restoration, under the supervision of Dr. Alper Gungor. Learnt the fundamentals of statistical signal processing.
 - Developed efficient super-resolution networks. Published CVPR and ECCV workshop papers as an undergraduate student.
- **Undergraduate Researcher** Ankara, Turkey
METU Center for Image Analysis (OGAM) 07/2021 - 06/2022
 - Worked on frame-based and event-based optical flow algorithms, under the supervision of Prof. Aydin Alatan.
 - Performed controlled hands-on experiments on a DAVIS event camera, testing its dynamic range and latency.
- **Part-Time Computer Vision Engineer** Ankara, Turkey
STM Inc. 08/2020 - 04/2021
 - Worked on real-time background subtraction, video dehazing, and object-tracking solutions for unmanned aerial vehicles (UAVs).
 - Refactored C++ code for real-time optimization, which is now used in the company's UAV products.

Honors & Scholarships

- Max Planck ETH Center for Learning Systems (*10 successful applicants over 1000*) and ETH AI Center doctoral fellow (2025-2029).
- Best CS MSc thesis award, given by the IEEE CS Turkey Branch (2025).
- Given merit scholarship by Bilkent University (*full tuition fee and monetary support, 2022-2025*).
- Given merit scholarship by METU Development Foundation (*top 0.04% in the entrance exam, 2018-2022*).
- Given merit scholarship by The Ministry of National Education of Turkey (2010-2018).

Selected Publications

- **B. Batuhan Bilecen**, Korrawe Karunratanakul, Vasileios Choutas, Thabo Beeler, Bernt Schiele, Jan Eric Lenssen, Siyu Tang. "InvAct: View and Scene-invariant Atomic Action Learning from Videos." In submission, 2026. ([🔗 project page](#))
- **A. Berke Gökmen, Yiğit Ekin, B. Batuhan Bilecen**, Ayşegül Dünder. "RoPECraft: Training-Free Motion Transfer with Trajectory-Guided Optimization on DITs." NeurIPS 2025. ([🔗 project page](#))
- **B. Batuhan Bilecen**, Yiğit Yalın, Ning Yu, Ayşegül Dünder. "Reference-Based 3D-Aware Image Editing with Triplanes". CVPR 2025 [Spotlight]. ([🔗 project page](#))
- **B. Batuhan Bilecen**, A. Berke Gökmen, Furkan Guzelant, Ayşegül Dünder. "Identity Preserving 3D Head Stylization with Multiview Score Distillation". ICCV 2025. ([🔗 project page](#))
- **B. Batuhan Bilecen, A. Berke Gökmen**, Ayşegül Dünder. "Dual Encoder GAN Inversion for High-Fidelity 3D Head Reconstruction from Single Images", NeurIPS 2024. ([🔗 project page](#))
- Alper Güngör, **B. Batuhan Bilecen**, Tolga Çukur. "Bayesian Conditioned Diffusion Models for Inverse Problems". In submission, 2024.
- Ahmet Burak Yıldırım, Hamza Pehlivan, **B. Batuhan Bilecen**, Ayşegül Dünder. "Diverse Inpainting and Editing with GAN Inversion", ICCV, 2023. ([🔗 project page](#))
- Alperen Kalay, **B. Batuhan Bilecen**, Mustafa Ayazoğlu. "Towards Clip-Free Quantized Super-Resolution Networks: How to Tame Representative Images", BMVC, 2023.
- **B. Batuhan Bilecen** and Mustafa Ayazoğlu. "Bicubic++: Slim, Slimmer, Slimmest - Designing an Industry-Grade Super-Resolution Network", NTIRE Workshop @ CVPR, 2023. [Challenge 1st place] ([🔗 project page](#))
- Mustafa Ayazoğlu and **B. Batuhan Bilecen**. "XCAT - Lightweight Quantized Single Image Super-Resolution Using Heterogeneous Group Convolutions and Cross Concatenation", AIM Workshop @ ECCV, 2022.
- **B. Batuhan Bilecen**, Alparslan Fişne, Mustafa Ayazoğlu. "Efficient Multi-Purpose Cross-Attention Based Image Alignment Block for Edge Devices", Embedded Vision Workshop @ CVPR, 2022.

University Projects

- **Building autonomous drones:** Built custom-frame and autonomous quad-copters with our student UAV Team, for the final rounds of 5th and 6th TUBITAK International Unmanned Aerial Vehicle Competitions, at TEKNOFEST. Learned the basics of *ArduPilot*, *MAVLink*, *DroneKit*, *ROS*, and *Gazebo*. Led the team with software setups. Modified ArduPilot C++ open source code to add our custom flight modes and controllers.
- **Signal processing:** Investigated SVD and applications to classical and deep priors in inverse problems in image restoration. [📄 \(doc link\)](#)
- **Neural Networks:** Derived the backpropagation for MLPs and RNNs, and implemented from scratch. ([🔗 project page](#))
- **Computer Architecture:** Designed a fully custom 16-bit instruction set architecture and a suitable multi-cycle CPU with *Verilog*. ([🔗 project page](#)) Designed a frequency-based motor controller with ARM Cortex M4. Wrote the code in *ARM Assembly*, using the Thumb 2 instruction set. ([🔗 project page](#)) Designed a point-of-sale device using Cyclone V FPGA. Utilized *Verilog* and wrote a VGA protocol handler from scratch. ([🔗 project page](#))

Skills & Interests

- **Languages:** Turkish (Native), English (Proficient), German (Beginner).
- **Hobbies:** Avid classical guitar player, currently learning Gran Vals by Tárrega. Enjoys reading about architecture, mobile photography, and taking nature walks.